



Skull Segmentation from CBCT Images via Voxel-Based Rendering

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Abstract. Skull segmentation from three-dimensional (3D) cone-beam computed tomography (CBCT) images is critical for the diagnosis and treatment planning of the patients with craniomaxillofacial (CMF) deformities. Convolutional neural network (CNN)-based methods are currently dominating volumetric image segmentation, but these methods suffer from the limited GPU memory and the large image size (*e.g.*, $512 \times 512 \times 448$). Typical ad-hoc strategies, such as down-sampling or patch cropping, will degrade segmentation accuracy due to insufficient capturing of local fine details or global contextual information. Other methods such as Global-Local Networks (GLNet) are focusing on the improvement of neural networks, aiming to combine the local details and the global contextual information in a GPU memory-efficient manner. However, all these methods are operating on regular grids, which are computationally inefficient for volumetric image segmentation. In this work, we propose a novel VoxelRend-based network (VR-U-Net) by combining a memory-efficient variant of 3D U-Net with a voxel-based rendering (VoxelRend) module that refines local details via voxel-based predictions on non-regular grids. Establishing on relatively coarse feature maps, the VoxelRend module achieves significant improvement of segmentation accuracy with a fraction of GPU memory consumption. We evaluate our proposed VR-U-Net in the skull segmentation task on a high-resolution CBCT dataset collected from local hospitals. Experimental results show that the proposed VR-U-Net yields high-quality segmentation results in a memory-efficient manner, highlighting the practical value of our method.

Keywords: VoxelRend · CBCT image · High-resolution segmentation

1 Introduction

Segmentation of high-resolution cone-beam computed tomography (CBCT) images plays an essential role in generating three-dimensional (3D) models in the diagnosis and treatment planning for patients with craniomaxillofacial (CMF)

deformities [13, 14]. Convolutional neural network (CNN)-based methods are currently dominating the 3D medical image segmentation. Among these CNNs, U-Net [12] is a breakthrough and has evolved to a commonly used benchmark in various segmentation challenges [1, 7, 11, 16]. However, CNN-based volumetric segmentation often requires massive memory of graphics processing units (GPUs) to process large image size dataset (e.g., millions of voxels). On the other hand, typical ad-hoc strategies, e.g., down-sampling [15] or patch cropping [4], can reduce the GPU memory usage but they also degrade segmentation quality due to insufficient capture of local fine details or global contextual information. Therefore, it is challenging to achieve a well-balanced trade-off between the segmentation accuracy and the GPU memory usage for CBCT images.

To tackle with this challenge, some related methods have been proposed in recent years. Data-swapping method [6] was proposed to swap the feature maps from GPU to central processing unit (CPU) memory. After the forward propagation process was finished, the feature maps were swapped back to GPU memory from CPU before executing the layer in the backward propagation process. However, this method requires additional overhead due to frequent swap computation. The fast segmentation methods like the Global-Local Network [2] is composed of a global branch and a local branch, taking the entire down-sampled image and its cropped local patches as respective inputs. With less than 2 GB GPU memory used, GLNet can yield decent segmentation results for ultra-high resolution images. The 3D Regions of Interest (RoI)-aware U-Net (3D RU-Net) [5] consists of a global image encoder and a GPU memory-efficient local decoder, performing a global-to-local multi-task learning procedure for joint RoI localization and in-region segmentation where the two tasks share one backbone encoder network. However, all the previous proposed neural networks operate on regular grids, which is convenient but not computationally efficient for image segmentation, in which most of the computation will unnecessarily focus on the low-frequency regions, e.g., the background, instead of the high-frequency regions, e.g., sparse boundaries between objects.

More recently, a method of treating image segmentation as a rendering problem was proposed in [9] for high-quality segmentation of 2D images. From this perspective, a fine-resolution segmentation can be rendered from a coarse-resolution segmentation by performing point-wise predictions on certain locations with high prediction uncertainties. In this work, we propose to extend this general perspective in the domain of the CBCT segmentation. Specifically, we propose a novel neural network, VoxelRend-based U-Net (VR-U-Net), which integrates a GPU memory-efficient variant of 3D U-Net with a voxel-based rendering (VoxelRend) module for high-resolution volumetric segmentation. The VoxelRend module is a flexible and slight-weight neural network derived from PointRend [9] to perform efficient voxel-wise predictions at challenging voxel locations. The proposed VR-U-Net was evaluated in the segmentation of multi-class bony structures on a high-resolution CMF CBCT dataset. It is challenging to segment the CBCT images due to their high-resolution (up to 100 million voxels) and the wild spread artifacts caused by noise, beam hardening, and

inhomogeneity. With only one single Titan XP GPU and less than 4 GB memory used, our VR-U-Net yields higher segmentation quality and achieves much more competitive accuracy memory usage trade-offs compared to state-of-the-arts. The experimental results have revealed that our proposed method has the huge potential for extension towards other volumetric segmentation tasks due to its inherent generalizability. Therefore, we make the code publicly available to encourage more researches on this topic.

2 3D VoxelRend-Based U-Net

In this section, we start by analyzing computation memory usage of the 3D U-Net [3] for binary segmentation with experimental statistics. Then, we present the network structure of the proposed VR-U-Net, which consists of a GPU memory-efficient variant of 3D U-Net serving as the backbone and a VoxelRend module established upon the backbone network. A brief description of the training and inference phase of the VR-U-Net is also described at the end of this section.

2.1 Memory Analysis of 3D U-Net

The memory issue of training and inference for 3D U-Net [3] is exacerbated by the encoder-decoder framework of 3D U-Net. The encoder part of 3D U-Net compresses the input image into low-resolution feature maps, while the decoder part decompresses these low-resolution feature maps until their original resolution is reached. In the training phase, all feature maps need to be retained for the backward computation. In the inference phase, the feature maps in the encoder part need to be retrained in the forward computation due to the skip connections between the encoder and decoder. Table 1 summarizes the experimental statistics on memory consumption of each layer of the 3D U-Net for binary segmentation. L-Stage 1 denotes the top layer of the encoder, and L-Stage 1 denotes the top layer of the decoder. From this table, we observe that the memory consumption of R-Stage 1 takes up more than one third of the overall memory consumption (marked in bold), and the memory consumption of R-Stage 1 together with that of the R-Stage 2 take up more than half of the overall memory consumption. Therefore, we build a GPU memory-efficient variant of 3D U-Net by discarding both the R-Stage 1 layer and R-Stage 2 layers in the 3D U-Net so that the R-Stage 3 layer is directly followed by the output layer, as shown in Fig. 1.

2.2 Network Architecture

Figure 1 shows the architecture of our proposed VR-U-Net, which consists of two modules. The first module is a GPU memory-efficient variant of 3D U-Net [3] serving as the backbone network, and the other one is a VoxelRend module established on the multi-layer feature maps of the backbone network. The backbone network is modified from 3D U-Net by discarding the top two layers of the decoder of 3D U-Net due to their heavy memory consumption.

Table 1. Summary of memory usage in each layer of 3D U-Net.

Layer	Input size	Output size	#Channels	#Conv.	Memory (%)
L-Stage 1	$128 \times 128 \times 64$	$64 \times 64 \times 32$	16	1	18.7
L-Stage 2	$64 \times 64 \times 32$	$32 \times 32 \times 16$	32	2	9.36
L-Stage 3	$32 \times 32 \times 16$	$16 \times 16 \times 8$	64	3	3.51
L-Stage 4	$16 \times 16 \times 8$	$8 \times 8 \times 4$	128	3	0.88
L-Stage 5	$8 \times 8 \times 4$	$16 \times 16 \times 8$	256	3	0.22
R-Stage 4	$16 \times 16 \times 8$	$32 \times 32 \times 16$	256	3	1.76
R-Stage 3	$32 \times 32 \times 16$	$64 \times 64 \times 32$	128	3	7.02
R-Stage 2	$64 \times 64 \times 32$	$128 \times 128 \times 64$	64	2	18.7
R-Stage 1	$128 \times 128 \times 64$	$128 \times 128 \times 64$	32	1	37.5
Output	$128 \times 128 \times 64$	$128 \times 128 \times 64$	2	1	2.34

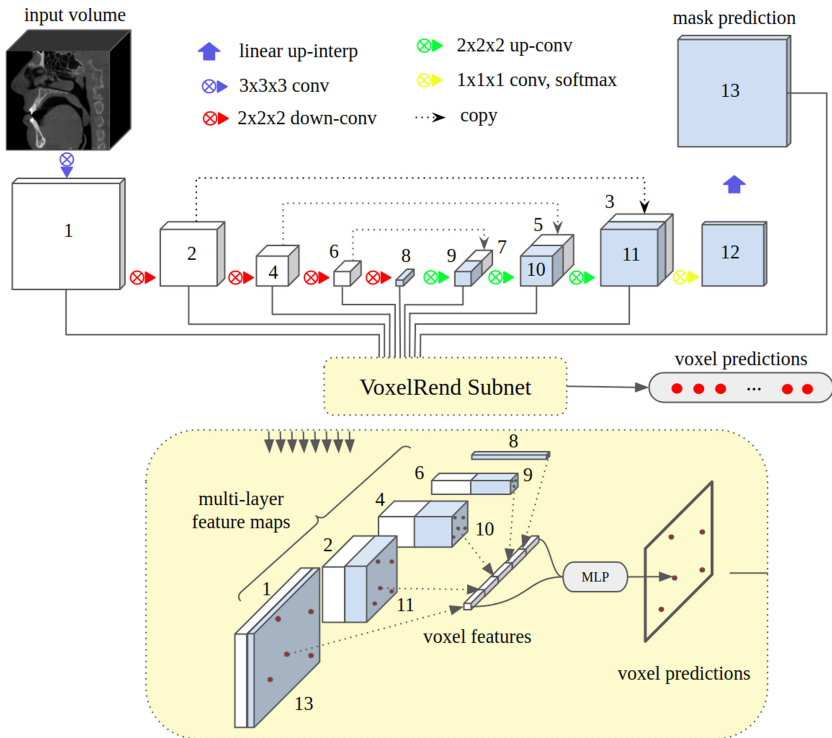


Fig. 1. Schematic representation of 3D VR-U-Net. The network consists of a backbone network and a VoxelRend module. The backbone network is a GPU memory-efficient variant of 3D U-Net that yields a mask prediction at a coarser resolution than the input volume. The VoxelRend module refines the coarse segmentation by performing voxel-wise predictions at adaptively selected locations (red dots) using a small MLP. The MLP uses interpolated features computed at these voxels (dashed arrows) from multi-layer feature maps generated by the backbone. (Color figure online)

Therefore, the output layer is directly following the third layer of the decoder part to produce the mask prediction at a relatively coarse resolution. The feature maps marked as 12 in the Fig. 1 represent the probability maps of the mask predictions. These probability maps will up-sampled to the original resolution using trilinear interpolation, which is memory-efficient due to no convolutional operations. The loss function of the backbone module is computed based on the up-sampled segmentation results. In order to improve the segmentation quality, the VoxelRend selects a set of voxels (red dots) and makes prediction for each voxel independently with a small multi-layer perceptron (MLP). This MLP uses interpolated features computed at these voxels (dashed arrows) from the multi-layer feature maps provided by the backbone network.

Similar to PointRend [9], the VoxelRend module consists of three components: 1) A *voxel selection sub-module* selects a few of real-value voxels to make predictions on. The voxels with higher prediction uncertainty (e.g., those with probabilities more closer to 0.5 for binary segmentation) are more likely to be selected. 2) A *voxel-wise feature extraction sub-module* computes feature representation for each selected voxel. The features for a real-value voxel is computed by trilinear interpolation of the feature maps. 3) A *voxel prediction sub-module* is trained to predict the class of all the selected voxels in one pass based on their feature representations.

2.3 Training and Inference

In the training phase, the backbone module and the VoxelRend module are learned jointly. The backbone module takes in the training volumes, and performs coarse mask prediction, which will be up-sampled to the original resolution for loss computation. The VoxelRend model performs voxel-wise predictions at locations where the predictions from the backbone module are ambiguous. For example, a voxel is more likely to be selected if its prediction probability is more closer to 0.5 for binary segmentation. Therefore, the loss function of the proposed VR-U-Net consists of the mask prediction loss L_m , and the voxel predictions loss L_v . The overall loss function L is the linear combination of L_m and L_v , which is formulated in Eq. 1.

$$L = \alpha L_m + (1 - \alpha) L_v, \alpha \in [0, 1] \quad (1)$$

Both L_m and L_v are implemented with Focal Loss [10] in this work. In the inference phase, the mask prediction is conducted first to obtain the coarse segmentation. Then, voxel predictions is performed on the N most uncertain voxels in the mask prediction. The number of N is a hyper-parameter. In our implementation, we select 5% of total voxels in a given volume.

3 Experiments and Results

3.1 Dataset

The effectiveness of the proposed VR-U-Net was evaluated using 92 sets of CBCT and 78 sets of CT images of normal subjects and patients with CMF deformities

(Table 2). The study was approved by Institutional Review Board, and all personal information was de-identified. Images were divided into 2 groups: 136 for training and 34 for test images using stratified sampling. All images in the dataset were acquired for orthographic surgical planning, and thus suffer from the clinical variability encountered in clinical settings. An example of the typical volume from the dataset is shown in Fig. 2.

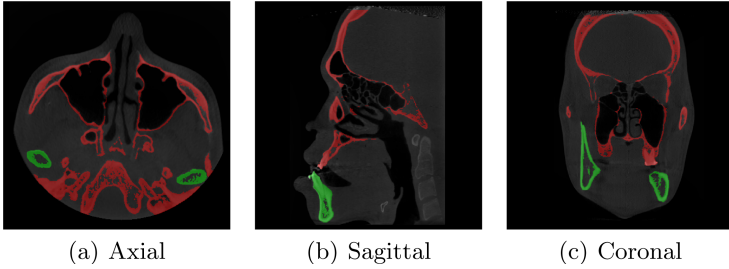


Fig. 2. An example CBCT image in our dataset with corresponding manual segmentation. Left: Axial view. Middle: Sagittal view. Right: Coronal view. The label in red is the midface, and the label in green is the mandible. (Color figure online)

Table 2. Summary of the dataset. All images were acquired once per subject.

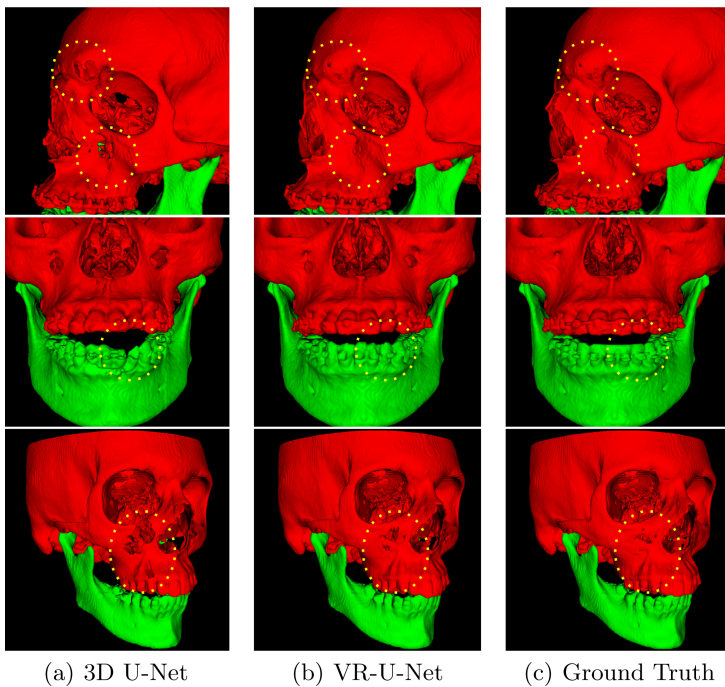
	Dataset
Number of scans	170
Subject type	Normal (26), patient (144)
Median spacing (mm ³)	$0.39 \times 0.39 \times 1.0$
Median resolution	$512 \times 512 \times 418$
Manual segmentations	Midface, mandible

3.2 Accuracy and Memory Usage Comparison

Table 3 shows the comparison results between the proposed VR-U-Net and the 3D U-Net [3] in terms of segmentation accuracy and memory usage in the inference phase. To investigate the effectiveness of the proposed VoxelRend module, we removed it from the VR-U-Net to obtain a degraded version of VR-U-Net for reference. The experimental results of this degraded VR-U-Net were also included in the Table 3. Our network enabled the largest input patch size of $224 \times 224 \times 224$, which was impractical for 3D U-Net [3]. To thoroughly compare the performance of our proposed method with the state-of-the-art 3D U-Net, we set different groups of experiment settings in terms of patch size and patch spacing.

Table 3. The comparison of accuracy and memory usage. The VoxelRend module is removed in the degraded VR-U-Net.

Model	Patch size	Spacing (mm ³)	Avg. Dice (%)		Memory (MB)
			Midface	Mandible	
3D U-Net	128 ³	0.4 ³	87.6 ± 7.1	89.4 ± 4.1	2607
	128 ³	0.8 ³	87.9 ± 6.4	89.3 ± 5.3	2607
	224 ³	0.4 ³	–	–	Out of memory
	224 ³	0.8 ³	–	–	Out of memory
Degraded VR-U-Net	128 ³	0.4 ³	84.8 ± 5.2	88.2 ± 5.9	1165
	128 ³	0.8 ³	84.4 ± 6.8	87.9 ± 5.4	1165
	224 ³	0.4 ³	85.0 ± 5.8	88.9 ± 5.7	3599
	224 ³	0.8 ³	84.7 ± 6.5	88.7 ± 6.4	3599
VR-U-Net	128 ³	0.4 ³	88.3 ± 6.6	92.2 ± 4.1	1165
	128 ³	0.8 ³	86.8 ± 4.0	90.8 ± 6.7	1165
	224 ³	0.4 ³	87.9 ± 4.8	92.4 ± 3.4	3599
	224 ³	0.8 ³	87.3 ± 6.4	91.3 ± 3.7	3599

**Fig. 3.** Illustration of the performance of VoxelRend module. Each row represents a test case. (a) Segmentation results of 3D U-Net. (b) Segmentation results of VR-U-Net. (c) Manual segmentation.

The Dice Similarity Coefficient (DSC) is used in our experiments as a measurement for the segmentation accuracy. The experimental results showed that by adding the VoxelRend module, the segmentation accuracy of VR-U-Net was improved. Figure 3 showed the improvements on three test cases. The experimental results also show that VR-U-Net achieved higher segmentation accuracy than the 3D U-Net using only half of the memory consumption compared with that of 3D U-Net.

Our experiments and models were implemented using Python 3.7 and Pytorch 1.4, with a standard workstation equipped with two Intel Xeon E5-2620 v4 CPUs working at 2.10 GHz, and a single NVidia Titan XP GPU with 12 GB memory. For each experiment, the model was trained for 2000 epochs using the Adam optimizer [8] with a momentum of 0.99 and an initial learning rate of 0.0001. We set the α in Eq. 1 to 0.5 by default. During each training iteration, the patches in the mini-batch were randomly cropped from the image domain with Z-score normalization. All inference was done patch-based as well that patches were cropped from the original image with no overlap. The average inference time for each experiment was less than 3 min.

4 Conclusion

In this work, we proposed a novel neural network for skull segmentation from high-resolution CBCT images. Our proposed network integrates a GPU memory-efficient variant of 3D U-Net with a VoxelRend module that performs voxel-based predictions on adaptively selected locations. We evaluated the proposed network on a challenging high-resolution CBCT image dataset. The experimental results show that our method yields high-quality segmentation results in a memory-efficient manner, highlighting the practical value of our method.

Acknowledgement. This work was supported in part by United States National Institutes of Health (NIH) grants R01 DE022676, R01 DE027251, and R01 DE021863.

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